ClipSAT : Egyptian Scholastic Test: EST II Mathematics Level 2 [ANSWER KEY]

Unit 3.01: Polynomial, Rational, Exponential & Logarithmic Functions

Q1. The graph of the function y = fx is shown below. What is the value of f1 ?

[x] A: 0

[ ] B: 2

[ ] C: -1

[ ] D: 3

Solution: The value of f1 corresponds to the y-coordinate of the point on the graph where the x-coordinate is 1. Looking at the provided graph, the point 1, 0 is marked. Therefore, f1 = 0 .

Q2. The graph of the function y = fx = ax^2 + bx + c is shown below. The graph passes through the points -1, 5 , 1, 1 , and 3, 5 . Determine the value of b .

[ ] A: 0

[x] B: -2

[ ] C: 2

[ ] D: 1

Solution: The general form of the quadratic function is y = ax^2 + bx + c . We are given three points that lie on the graph: -1, 5 , 1, 1 , and 3, 5 . Substituting these points into the equation, we get a system of three linear equations with three unknowns a, b, and c: 1. For -1, 5 : a-1^2 + b-1 + c = 5 Rightarrow a - b + c = 5 2. For 1, 1 : a1^2 + b1 + c = 1 Rightarrow a + b + c = 1 3. For 3, 5 : a3^2 + b3 + c = 5 Rightarrow 9a + 3b + c = 5 We can solve this system of equations. Let's subtract equation 1 from equation 2: a + b + c - a - b + c = 1 - 5 2b = -4 b = -2 Alternatively, notice that the y-values for x = -1 and x = 3 are the same both are 5. This indicates that the axis of symmetry of the parabola is exactly in the middle of x = -1 and x = 3 . The x-coordinate of the vertex axis of symmetry is given by -frac{b}{2a} . Axis of symmetry: x = frac{-1 + 3}{2} = frac{2}{2} = 1 So, -frac{b}{2a} = 1 , which implies b = -2a . Now, substitute this into equation 2: a + b + c = 1 a + -2a + c = 1 -a + c = 1 Rightarrow c = a + 1 Substitute b = -2a and c = a + 1 into equation 1: a - -2a + a + 1 = 5 a + 2a + a + 1 = 5 4a + 1 = 5 4a = 4 a = 1 Now we can find b using b = -2a : b = -21 b = -2 To confirm, we can also find c : c = a + 1 = 1 + 1 = 2 . The function is y = x^2 - 2x + 2 . Let's check the points: For -1, 5 : -1^2 - 2-1 + 2 = 1 + 2 + 2 = 5 Correct For 1, 1 : 1^2 - 21 + 2 = 1 - 2 + 2 = 1 Correct For 3, 5 : 3^2 - 23 + 2 = 9 - 6 + 2 = 5 Correct The value of b is -2.

Q3. Consider the functions fx = x^2 - 6x + 5 and gx = -x + 5 . The graphs of these two functions intersect at two points. Let these points be A and B . A third function, hx , is defined as the difference between the x-coordinates of these intersection points, i.e., hx = |x_A - x_B| . What is the value of hx ?

[ ] A: sqrt{16}

[x] B: sqrt{25}

[ ] C: sqrt{36}

[ ] D: sqrt{49}

Solution: To find the intersection points of the graphs of fx and gx , we set fx = gx . x^2 - 6x + 5 = -x + 5 Subtract -x + 5 from both sides: x^2 - 6x + 5 - -x + 5 = 0 x^2 - 6x + 5 + x - 5 = 0 Combine like terms: x^2 - 5x = 0 Factor out x : xx - 5 = 0 This gives two possible x-values for the intersection points: x_1 = 0 x_2 = 5 Let x_A = 0 and x_B = 5 or vice versa. The problem defines hx as the absolute difference between these x-coordinates: hx = |x_A - x_B| . hx = |0 - 5| = |-5| = 5 . We are asked for the value of hx , which is 5. We need to express this value in a form that matches the choices. The choices are given as square roots. sqrt{16} = 4 sqrt{25} = 5 sqrt{36} = 6 sqrt{49} = 7 The value of hx is 5, which corresponds to sqrt{25} .

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Q4. Consider the function fx = x^2 - 4x + 3 . What are the coordinates of the vertex of the parabola represented by this function?

[x] A: 2, -1
[ ] B: 2, 1
[ ] C: 1, 0
[ ] D: 3, 0

Solution: The given function is a quadratic function in the form fx = ax^2 + bx + c , where a=1 , b=-4 , and c=3 . The x-coordinate of the vertex of a parabola is given by the formula x = -frac{b}{2a} . Substituting the values, we get x = -frac{-4}{21} = frac{4}{2} = 2 . To find the y-coordinate of the vertex, we substitute this x-value back into the function: f2 = 2^2 - 42 + 3 = 4 - 8 + 3 = -1 . Therefore, the coordinates of the vertex are 2, -1.

Q5. The graph of the quadratic function fx = x^2 + bx + c is shown below. The graph passes through the point -2, 5 and has its vertex at 1, -4 . Determine the value of b .

[ ] A: 1
[x] B: -2
[ ] C: -1
[ ] D: 2

Solution: The vertex of a parabola given by fx = ax^2 + bx + c is at x = -frac{b}{2a} . In this case, a = 1 and the x-coordinate of the vertex is given as 1. So, 1 = -frac{b}{21} . This simplifies to 1 = -frac{b}{2} , which means b = -2 . Alternatively, we can use the vertex form of a quadratic equation: fx = ax-h^2 + k , where h, k is the vertex. Here, a=1 and h, k = 1, -4 , so fx = x-1^2 - 4 . Expanding this, we get fx = x^2 - 2x + 1 - 4 = x^2 - 2x - 3 . Comparing this to fx = x^2 + bx + c , we see that b = -2 and c = -3 . We can verify that the point -2, 5 lies on this parabola: f-2 = -2^2 - 2-2 - 3 = 4 + 4 - 3 = 5 . This confirms our values. Common errors might involve misinterpreting the vertex formula or making sign errors during expansion.

Q6. The graph of the function fx = -x^2 + 4x - 3 is shown below. Determine the y-coordinate of the vertex of the parabola.

[x] A: 1
[ ] B: 2
[ ] C: 3
[ ] D: 4

Solution: The vertex of a parabola in the form fx = ax^2 + bx + c occurs at x = -frac{b}{2a}. In this case, a = -1 and b = 4. So, the x-coordinate of the vertex is x = -frac{4}{2-1} = -frac{4}{-2} = 2. To find the y-coordinate of the vertex, substitute this x-value back into the function: f2 = -2^2 + 42 - 3 = -4 + 8 - 3 = 1 . Therefore, the vertex is at 2, 1 and the y-coordinate is 1.

Q7. The graph of the quadratic function fx = ax^2 + bx + c is shown below. The vertex of the parabola is at 2, -5 and the y-intercept is at 0, -1 . Determine the value of a .

[x] A: 1
[ ] B: -1
[ ] C: 2
[ ] D: -2

Solution: The general form of a quadratic function is fx = ax^2 + bx + c . The vertex form of a quadratic function is fx = ax-h^2 + k , where h, k is the vertex. Given the vertex is 2, -5 , the function can be written as fx = ax-2^2 - 5 . We are also given that the y-intercept is 0, -1 . This means that when x=0 , fx=-1 . Substituting these values into the vertex form: -1 = a0-2^2 - 5 . This simplifies to -1 = a-2^2 - 5 , which is -1 = 4a - 5 . Adding 5 to both sides gives 4 = 4a . Dividing by 4 yields a = 1 .

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Q8. Consider the graph of the function fx shown below. What is the value of f0 ?

[x] A: -1

[ ] B: 1

[ ] C: 0

[ ] D: 2

Solution: The value of f0 is the y-coordinate of the point where the graph of the function intersects the y-axis. From the graph, we can see that the function intersects the y-axis at the point 0, -1 . Therefore, f0 = -1 .